INCH-POUND

MIL-STD-1860

21 December 1988

MILITARY STANDARD

BATTERIES, NON-RECHARGEABLE

SELECTION AND USE OF



AMSC N/A

FSC 6135

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:26Z
Check the source to verify that this is the current version before use.

FOREWORD

1. This military standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, LABCOM, Attn: SLCET-RS, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-24T01:26Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		Page
1.	SCOPE	1
1.1	Scope	1
1.2	Purpose of Standard	1
2.	REFERENCED DOCUMENTS	2
2.1	Government documents	2
2.1.1	Specifications	2
2.1.2	Other Government documents	2
2.2	Order of precedence	2
3.	DEFINITIONS	3
3.1	Battery	3
3.1.1	Standard battery	3
3.1.2	Non-rechargeable battery	3
3.2	Standard voltage	3
4.	GENERAL REQUIREMENTS	4
4.1	Standard types	4
4.2	Assignment of non-rechargeable battery types	4
4.2.1	Non-standard batteries	4
5.	DETAILED REQUIREMENTS	5
5.1	Characteristics of standard non-rechargeable batteries	5
6.	NOTES	7
6.1	Intended use	7
6.2	Subject term (key word) listing	7
6.3	Application notes	7
6.3.1	Lithium batteries	7
6.3.2	Alkaline batteries	7
6.4	Storage information	7
6.4.1	Lithium batteries	7
6.4.2	Alkaline batteries	7
6.5	Transportation of lithium batteries	7
6.6	Disposal information	7
6.6.1	Lithium batteries	8
6.6.2	Alkaline batteries	8

TABLE		
I	Alkaline batteries	6
II	Lithium batteries	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:26Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard consists of selected standard non-rechargeable battery types for use in military equipment. It provides general information for the choice and application of non-rechargeable batteries. Detailed requirements for non-rechargeable batteries listed herein are covered in the applicable specification (see 2.1).

1.2 Purpose of standard. This standard provides the equipment designer with a selection of standard non-rechargeable batteries for use in military equipment. It is intended to minimize and control the variety of non-rechargeable batteries used in military equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:26Z
Check the source to verify that this is the current version before use.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

MILITARY SPECIFICATIONS

MIL-B-49030(ER)	Batteries, Dry, Alkaline
MIL-B-49430(ER)	Batteries, Non-Rechargeable, Lithium Sulfur Dioxide
MIL-B-49458(ER)	Batteries, Non-Rechargeable, Lithium Manganese Dioxide

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this standard to the extent specified herein. Unless otherwise specified, the issues are those listed in the solicitation.

DOT-E-7052	Transportation Exemption for Lithium Batteries
AFR 71-4	Preparation of Hazardous Materials for Military Air Shipment

(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2. Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 202610-24T01:26Z
Check the source to verify that this is the current version before use.

3. DEFINITIONS

3.1 Battery. A portable electrochemical power source consisting of one or more electrochemical power cells with all necessary connectors, fusing, thermal protection devices, and wiring which are encased in a jacket to provide electrical power.

3.1.1 Standard battery. A battery of standard size and voltage that meets an applicable military specification for batteries including material of construction and safety features.

3.1.2 Non-rechargeable battery. A battery which derives its energy from an essentially irreversible electrochemical reaction and which is hence incapable of being efficiently recharged by an electric current.

3.2 Standard voltage. Standard voltages are those voltages most commonly used to power battery operated equipment. For the purpose of this standard, they include but are not limited to the following: 1.5, 3.0, 6.0, 9.0, 12.0, and 24.0 volts DC. These are the voltages under load. Standard voltage is also referred to as nominal voltage.

Source: https://assist.dla.mil -- Downloaded: 202640-24T01:26Z
Check the source to verify that this is the currentversion before use.

4. GENERAL REQUIREMENTS

4.1 Standard Types. Only non-rechargeable battery types listed in this standard shall be used in non-rechargeable battery operated equipment unless it is demonstrated that:

(a) Batteries listed herein cannot meet the equipment power requirements.

(b) Batteries not listed herein meet or exceed power and safety requirements of the equipment, while providing at the same time tactical, economic, and logistic advantages over the batteries listed herein.

4.2 Assignment of non-rechargeable battery types.

4.2.1 Non-standard batteries. The assignment of batteries not listed herein for military application shall be approved by the contracting agency. Request for approval to use batteries not listed herein shall be forwarded to military parts control advisory group (MPCAG) and the contracting agency. It shall be accompanied by written justification for selection of a non-standard battery.

Source: https://assist.dla.mil -- Downloaded: 202610-24T01:26Z
Check the source to verify that this is the current version before use.

5. DETAILED REQUIREMENTS

5.1 Characteristics of standard non-rechargeable batteries. General physical and electrical characteristics of the standard non-rechargeable batteries are shown in Tables I and II. Detailed physical and electrical characteristics can be found in the applicable specification.

Source: https://assist.dla.mil -- Downloaded: 202610-24T01:26Z
Check the source to verify that this is the current version before use.

TABLE I - Alkaline batteries.

MIL-B-49030 Spec Sheet	Battery Nomen-clature	Nominal Voltage (Volts)	Cell Size	Height 1/	Width 1/ or Diameter	Length 1/	Weight (Lbs) (Max)	Capacity (Amp-Hrs) 2/
/1	BA-3030/U	1.5	D	2.41	1.34	--	0.34	8.0
/3	BA-3042/U	1.5	C	1.97	1.03	--	0.15	3.56
/6	BA-3058/U	1.5	AA	1.97	0.58	--	0.05	1.25
/11	BA-3090/U	9.0	M2200	1.91	0.66	1.03	0.15	0.50

TABLE II - Lithium batteries.

Specification Number	Battery Nomenclature	Nominal Voltage (Volts)	Height 1/	Width 1/ or Diameter	Length 1/	Weight (LBS) (Max)	Capacity (Amp-Hrs) 2/
MIL-B-49458/1	BA-5372/U	6	1.32	0.66	----	0.0438	0.45
MIL-B-49430/6	BA-5557/U	12/24	3.94	1.48	4.18	1.1	2.25/4.50
MIL-B-49430/4	BA-5567/U	3	0.75	1.06	----	0.04	0.85
MIL-B-49430/7	BA-5588/U	12	3.76	1.24	3.56	0.65	2.90
MIL-B-49430/3	BA-5590/U	12/24	5.00	2.45	4.40	2.25	6.90/13.80
MIL-B-49430/2	BA-5598/U	12	2.06	3.56	4.69	1.5	7.20
MIL-B-49430/1	BA-5599/U	9	2.50	1.50	6.00	1.0	6.90
MIL-B-49430/8	BA-5600/U	9	7.34	1.56	----	0.8	6.00
MIL-B-49430/11	BA-5800/U	6	5.06	1.40	----	0.48	6.90
MIL-B-49430/5	BA-5847/U	6	3.78	1.53	2.58	0.53	6.90

1/ All dimensions are maximum and are in inches.

2/ Capacity is measured at 25°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:26Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard is intended to be used by the equipment designer to assist in the selection of a standard non-rechargeable battery.

6.2 Subject term (key word) listing.

Alkaline
Battery
Cell
Lithium
Manganese Dioxide
Non-rechargeable
Sulfur Dioxide

6.3 Application notes.

6.3.1 Lithium batteries. These batteries shall be used for applications requiring high rate of discharge (i.e. greater than 150 mA) and for low temperature use. Operating temperature range is -28.9°C to +55°C.

6.3.2 Alkaline batteries. These batteries shall be used for low to moderate rate of discharge (i.e. less than 150 mA). Operating temperature range is -17.8°C to +55°C.

6.4 Storage information. Batteries shall not be stored near heat or open flames.

6.4.1 Lithium batteries. Lithium batteries should be used within five years of date of manufacture. These batteries shall be stored in a well ventilated area between temperatures of 0°C and 55°C.

6.4.2 Alkaline batteries. Alkaline batteries shall be refrigerated, if possible. These batteries should be used within three years of date of manufacture.

6.5 Transportation of lithium batteries. Commercial transportation of these batteries shall conform to DOT-E-7052. Military air transportation shall conform to AFR 71-4.

Source: https://assist.dla.mil -- Downloaded: 202610-24T01:26Z
Check the source to verify that this is the current version before use.

6.6 Disposal information.

6.6.1 Lithium batteries. Upon completion of use, lithium batteries containing a complete discharge device may be disposed as normal trash after its discharge switch is activated. If the battery is not fully discharged, wait five days after activating the switch before discarding. Upon completion of use, lithium batteries not containing a complete discharge device shall be turned in for disposal to the regional Defense Reutilization Marketing Office (DRMO). Lithium batteries containing less than 0.5 grams of lithium (BA-5372/U and BA-5567/U) shall be excluded from the above policy. These lithium batteries containing less than 0.5 grams may be disposed of as normal trash.

6.6.2 Alkaline batteries. Alkaline batteries may be disposed of as normal trash.

Custodians:
Army-ER
Navy-SH
Air Force -

Review Activity:
Army - AR, AT, CR
Navy - EC
Air Force -
DLA-GS

User Activity:
Army-ME
Navy -
Air Force - 85

Preparing Activity:
Army-ER

(Project 6135-0206)

Source: https://assist.dla.mil -- Downloaded: 202610-24T01:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STD-1860	Batteries, Non-Rechargeable Selection and Use of

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:26Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

US ARMY ELECTRONICS TECHNOLOGY
& DEVICES LABORATORY (LABCOM)
FORT MONMOUTH, NJ 07703-5302
SLCET-R

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 Alexandria, VA

POSTAGE WILL BE PAID BY

US ARMY ELECTRONIC TECHNOLOGY
& DEVICES LABORATORY (LABCOM)
ATTN: SLCET-RS
FORT MONMOUTH, NJ 07703-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:26Z
Check the source to verify that this is the current version before use.